

Theoretical solvent recovery and grafting application in Xinjiang waste PE based on solubility parameters

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Research Article

Keywords: PE, Solvent Screening, Grafting, HSPiP , Interaction parameters

Posted Date: January 23rd, 2024

DOI: <https://doi.org/10.21203/rs.3.rs-3804093/v1>

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Abstract

This paper investigates the recovery of waste PE (mulch film) from Xinjiang by solvent recovery method. Under the guidance of solubility parameter theory, combined by HSPiP version 4.1.07 experiment and HSPs library screening, xylene was selected as the most suitable solvent for recovery of PE. Subsequently, the solid-liquid method was used to graft MAH to PE, and the optimal process conditions were obtained through orthogonal experiments, and the successful grafting of MAH to PE was verified by infrared spectroscopy. Then through XRD and other checks to analyze the PE before and after grafting, MAH did not change the crystal arrangement of PE, nor did it alter the crystal type, the introduction of MAH enhanced the surface energy of PE and reduced the regularity of the PE structure. The results of DSC showed that after MAH grafting, the nucleation site on the surface of PE increased, which could make the modified PE crystallize rapidly, so that the cold crystallization peak disappeared, and the stability was enhanced, which is of great significance for subsequent asphalt modification.

Introduction

The annual use of agricultural PE materials in China is about 4000 kt, of which PE mulch film alone reaches 1500 kt. A huge amount of agricultural PE residual film (including recycled PE mulch film residue, shed film residue, drip irrigation pipe, etc.) has formed a serious plastic pollution problem, and the amount of waste film accumulated in the field without treatment in Xinjiang is as high as one million tons [1]. In 2018, the "Agricultural Film Recycling Action Plan" was released by the Ministry of Agriculture. Additionally, China places great importance on reducing the pollution caused by used agricultural film, the safe recycling of waste plastics was identified as a pivotal component of the national strategic emerging industry by the National Development and Reform Commission in 2017[2]. With the implementation of the national green development strategy, there is an increasing demand for recycling of agricultural waste and waste separation, the agricultural residual film recycling system is basically established, and now the most critical problem is how to resource the recycled residual film, especially for the resource utilization of waste film mixed with stones, soil and even dead leaves. The traditional methods of waste PE treatment are mechanical recycling, chemical recycling and biodegradation. With the progress of technology and policy support, PE recycling will be more widely used in the future, contributing to the construction of a green society and promote the goal of sustainable development.

Maleic anhydride (MAH) is a small monomer containing polyfunctional groups whose double bonds can polymerize with polymers under initiator or irradiation conditions, and acid anhydride groups can react with groups such as amino groups, hydroxyl groups, and carboxyl groups [3]. Due to the symmetrical structure of MAH monomers, the auto polymerization of MAH is more difficult, and a long branched chain structure will not be formed on PE during the grafting process, thus avoiding a sharp decline in the mechanical properties of PE [4]. At present, there have been many studies on modified LDPE by MAH grafting at home and abroad, and the main methods include solution grafting [5], melt grafting [6–8] and solid phase grafting [9]. Many researchers use melt grafting to modify PE. Iqbal et al.[7]prepared LDPE-g-

MAH by twin screw extrusion, and determined that MAH was successfully grafted onto the molecular chain of LDPE by infrared characterization.

This essay will utilize the principles of solubility parameters theory to incorporate the findings from solubility experiments into the HSP (Hansen Solubility Parameters) database. Subsequently, it will involve the computation of the three-dimensional solubility parameters of PE and the derivation of their interaction parameters, leveraging thermodynamic properties. On the other hand, the solvent screening results are verified according to the HSPiP experimental results. Finally, according to the results of HSPiP experiments and HSPs theory, xylene was screened as a solvent. Graft modification of PE using the solvent method. The optimal grafting process conditions were obtained by orthogonal experimental analysis, after completing the grafting experiment, verify whether grafted monomer is grafted to PE by FT-IR. Characteristics such as thermal stability and enthalpy of melting of PE before and after modification were investigated by DSC, and XRD was allowed to determine whether the introduction of MAH monomer would change the crystal structure and crystal morphology of PE.

1. Experimental Procedure

1.1 Materials

Agricultural PE, Purchased from Outdoor farmland in Xinjiang. Xylene was purchased from Tianjin Xinbote Chemical Co., Ltd. DCP and MAH were both purchased from Aladdin Reagents. Acetone (analytical pure, Purity: 0.995), purchased from Chengdu Cologne Chemical Co., Ltd.

1.2 Selection of solvent

Based on the basic principle of "similar compatibility", HSPiP calculates the three-dimensional solubility parameter of the material to be measured from the actual dissolution between the solvent and the material to be measured, and it can be calculated by the genetic algorithm of the material to be tested. Theoretically, the more types of solvents used in the simulation, the more accurate the simulation results [10]. The specific operation process is shown in Fig. 1.

1.2.1 Solubility test of PE

According to the existing 63 organic solvents in the laboratory, the PE mixed with organic solvents, and then the solubility is observed by HSPiP experiment, after that, enter the dissolution into the HSPiP software, in which to deliberate the three-dimensional solubility parameters of PE. The obtained solubility data for various solvents and their interaction with PE are inputted into the HSPiP software, subsequently enabling the generation of a three-dimensional solubility sphere that represents the compatibility of these solvents with PE. Additionally, to guarantee the experiment's accuracy, there are two experiments were made for control. As shown in Fig. 2, the green mesh ball represented the three-dimensional solubility ball of PE, and the blue solid ball means its good solvent of PE, located inside the

three-dimensional solubility ball. On the contrary, the red solid ball is a bad solvent of PE, located in the interior of the three-dimensional solubility ball, the green solid ball represents PE, located at the center position of the three-dimensional solubility ball, the spatial three-dimensional coordinates are δ_D , δ_P and δ_H respectively, the detailed values are listed in Table 1.

1.2.2 Screening of HSPs libraries

First of all, we obtained the three-dimensional solubility parameters of PE based on the previous experiments, and then input them into the HSPs library to fit them, according to the solubility parameter theory to screen out the solvent with the RED value close to it, in general, the RED value less than 1 is a good solvent, otherwise it is a bad solvent, the smaller the RED value, the more effectively these solvents to dissolve PE, the RED value of each solvent is shown in Table 1.

Table 1
Solvent screening results for PE

Solvent name	Boiling(°C)	Three-dimensional solubility parameters				RED	Dissolution with PE	X_{HSP}
		δ_D	δ_P	δ_H	δ_t			
pentane	36	14.10	0.00	0.00	14.10	1.26	0	1.66
n-Hexane	69	14.90	0.00	0.00	14.90	0.53	1	1.21
n-Heptane	98	15.30	0.00	0.00	15.30	0.47	1	1.01
n-octane	125–127	15.50	0.00	0.00	15.50	1.18	0	0.98
Nonane	151	15.70	0.00	0.00	15.70	1.16	0	0.95
N-Decane	174	15.70	0.00	0.00	15.70	1.16	0	1.04
Undecane	196.28	16.00	0.00	0.00	16.00	1.138	0	0.93
Dodecane	216.3	16.00	0.00	0.00	16.00	1.138	0	1.00
cyclopentane	49.2	16.40	0.00	1.80	16.50	1.05	0	0.01
Cyclohexane	80.7	16.80	0.00	0.20	16.80	0.29	0	0.30
Isooctane	98–99	14.10	0.00	0.00	14.10	1.25	0	2.41
benzene	80.1	18.40	0.00	2.00	18.51	0.44	2	0.41
toluene	110.6	18.00	1.40	2.00	18.16	0.26	2	0.23
ethylbenzene	136.2	17.80	0.60	1.40	17.87	0.25	2	0.09
xylene	137–140	17.60	1.00	3.10	17.90	0.16	2	0.10
o-Xylene	143–145	17.40	1.90	2.70	17.71	0.64	1	0.10
m-Xylene	139	18.00	2.30	2.30	18.29	0.46	2	0.34
p-Xylene	138.4	17.80	1.00	3.10	18.10	0.89	1	0.15
1-Octene	122–123	15.30	1.00	2.40	15.52	0.63	1	0.61
1-Hexene	62.8	14.70	1.10	3.00	15.04	1.17	0	0.90
Cyclohexene	85.6	17.20	1.00	2.00	17.34	0.98	1	0.04
styrene	145.2	18.60	1.00	4.10	19.07	0.81	1	0.58
methanol	64.8	14.70	12.30	22.30	29.41	1.26	0	8.09
ethanol	78.3	15.80	8.80	1.70	18.17	1.04	0	0.91

Solvent name	Boiling(°C)	Three-dimensional solubility parameters				RED	Dissolution with PE	X_{HSP}
		δ_D	δ_P	δ_H	δ_t			
n-Propanol	97.4	16.00	6.80	17.40	24.60	0.97	1	7.16
isopropanol	82.5	15.80	6.10	16.40	23.58	0.96	1	6.22
Butanol	117.6	16.00	5.70	15.80	23.20	1.14	0	6.79
sec-butanol	99.5	15.80	5.70	14.50	22.19	0.95	1	5.76
Isobutanol	105	15.10	5.70	15.90	22.66	1.03	0	7.24
tert-Butanol	84.6	15.20	5.10	14.70	21.75	1.01	0	5.96
ethylene glycol	197.3	17.00	11.00	26.00	32.95	0.63	1	12.40
glycerol	290	17.40	11.30	27.20	34.21	0.64	1	18.83
Pentyl alcohol	137–139	15.90	5.90	13.90	21.93	0.95	1	15.70
Octanol	196	16.00	5.00	11.20	20.16	0.935	1	5.43
formamide	210.5	17.20	26.20	19.00	36.65	1.52	0	15.70
ethanolamine	170.9	17.00	15.50	21.00	31.15	1.15	0	12.90
Triethanolamine	335.4	17.30	7.60	21.00	28.25	1.22	0	19.16

Continued Table 1 Solvent screening results for PE

Solvent name	Boiling(°C)	Three-dimensional solubility parameters				RED	Dissolut ion with PE	X _{HSP}
		δ_D	δ_P	δ_H	δ_t			
Cyclobutyl sulfone	285	17.80	17.40	8.70	26.39	0.83	1	11.22
N,N-Dimethylformamide	152.8	17.40	13.70	11.30	25.22	1.01	0	7.44
N,N-Dimethylacetamide	164.5–166	16.80	11.50	9.40	22.42	1.01	0	5.88
Methyl formate	31.5	15.30	8.40	10.20	20.22	1.06	0	3.01
Methyl acetate	56.8	15.50	7.20	7.60	18.70	1.12	0	2.20
ethyl acetate	77	15.80	5.30	7.20	18.15	0.72	1	1.64
Methyl propionate	79.8	15.50	5.50	6.70	17.76	1.04	0	1.66
Ethyl propionate	99	15.50	6.10	4.90	17.36	1.09	0	1.74
Ethyl butyrate	122.4	15.50	5.50	5.00	17.19	1.36	0	2.71
Butyl ether	142–143	15.20	3.40	3.20	15.90	1.13	0	1.16
Propylene oxide	32.24	15.20	8.60	6.70	18.71	0.62	1	2.11
Tetrahydrofuran	66	16.80	5.70	8.00	19.46	0.47	2	1.53
thiophene	84.2	18.90	2.40	7.80	20.59	1.29	0	1.44
pyridine	115	19.00	8.80	5.90	21.75	1.12	0	2.83
N-Methylpyrrolidone	202	18.00	12.30	7.20	22.96	1.27	0	5.66
Dimethyl sulfoxide	189	18.40	16.40	10.20	26.68	1.04	0	9.52
1,4-Dioxane	101.1	17.50	1.80	9.00	19.76	0.49	2	1.36
Acetonitrile	81.6	15.30	18.00	6.10	24.40	1.36	0	6.55
Cyclohexanone	155.6	17.80	8.40	5.10	20.33	0.88	1	2.57
2-Pentanone	101–105	16.00	7.60	4.70	18.33	1.07	0	2.95
3-Pentanone	101	15.80	7.60	4.70	18.15	1.08	0	2.22
dichloromethane	39.8	17.00	7.30	7.10	19.82	0.84	1	1.55
carbon tetrachloride	76.7	17.80	0.00	0.60	17.81	1.18	0	0.32
Dimethyl carbonate	165	15.50	8.60	9.70	20.21	1.05	0	3.84
carbon disulfide	46.3	19.90	5.80	0.60	20.74	1.12	0	1.49

Solvent name	Boiling(°C)	Three-dimensional solubility parameters				RED	Dissolut ion with PE	X_{HSP}
		δ_D	δ_P	δ_H	δ_t			
ligroin	40–80	17.90	0.70	1.80	18.00	0.93	1	

1.2.3 Selection of solvent based on the thermodynamic properties of PE

The HSP-based interaction parameters (X_{HSP}) are calculated by the following formula:

$$X_{HSP} = \frac{V_1}{RT} \left[(\delta_{1,d} - \delta_{2,d})^2 + 0.25 (\delta_{1,p} - \delta_{2,p})^2 + 0.25 (\delta_{1,h} - \delta_{2,h})^2 \right]$$

1

In this equation, V_1 represents the molar volume of the organic solvent, while $\delta_{1,d}$, $\delta_{1,p}$, and $\delta_{1,h}$ denote the three-dimensional solubility parameters of component 1. Similarly, $\delta_{2,d}$, $\delta_{2,p}$, and $\delta_{2,h}$ represent the corresponding solubility parameters for component 2.

In general, the ability of two substances to interact with each other can be reflected by the value of the Flory-Huggins interaction parameter. The smaller the value of the interaction parameter between the two substances, the stronger the interaction between the two substances, and on the contrary, the larger the value, the smaller the interaction relationship [11].

According to the X_{HSP} values in Table 1, cyclopentane, cyclohexane, ethylbenzene, benzene, toluene, xylene, o-xylene, p-xylene, cyclohexene, styrene and carbon tetrachloride was used as a good solvent, xylene was chosen as the solvent for the grafting experiments based on the RED value and industrial price as well as toxicity and other performance results.

2. Grafting modification of MAH

Figure 3 shows the mechanism of the grafting reaction, DCP is heated to seize the hydrogen atoms on the chain of PE macromolecules, MAH reacts with it to generate macromolecular radicals, and then reacts with PE to form grafting products as well as new PE macromolecular radicals.

2.1 Preparation of PE-g-MAH

In a four-mouth flask, nitrogen is introduced at one end, a thermometer is inserted at one end, a condenser tube at one end, and a constant pressure separating funnel at the other end. Add a certain amount of waste PE and xylene solution in order to completely dissolve the PE in an oil bath while heating and stirring [16]. Gradually heat while stirring to the designated reaction temperature under the protection of N_2 . Subsequently, add a specific dose of MAH. Add the DCP into the constant pressure separating funnel

for titration and record the start time. Then, add drops of DCP while measuring the total drop-adding time at the end of the experiment. Ensure to abide by appropriate lab protocols during the experiment [12]. Reaction ends when reaction time is reached. When the temperature is cooled to 70–80°C, gently poured the acetone solution into the product at an ice water bath, then followed by stirring, soaking, precipitation. Finally, vacuum filtration of the product, repeatedly cleaning with acetone three times [13]. The product is a primary product after 4 hours drying by a vacuum oven, which may contain some unfinished maleic anhydride, residual initiator, dimers and other impurities. Thus, need to be further purified.

2.2 Purification of PE-g-MAH

The dried graft product and xylene solvent were combined in a flask for a condensation reflux setup. The mixture was then heated to reach a reflux state in a water bath, and this reflux process was maintained for three hours to eliminate any remaining unreacted impurities. The purification operation was performed as before.

The initially purified product is allowed to sit for 4 hours to eliminate any unreacted components and unwanted impurities, and then the refined product is obtained after vacuum drying.

2.3 Determination of grafting rate

The grafting rate primarily refers to the mass or molar ratio of the grafted MAH to the grafted polymer [14]. There is no doubt that using the back titration method is very suitable, this method can make the grafted PE on maleic anhydride and alkali full reaction complete, that is, first in a certain temperature and solvent, let the grafted product completely dissolve, and then add a certain excess of standard alkali solution, so that the acid anhydride and lye fully react. Subsequently, employ the acid solution for an indirect titration method to ascertain the concentration of the alkali solution. This, in turn, allows for the indirect determination of the acid anhydride content in the target product [15]. Configure standard acid-base solutions for titration and determine the grafting rate of the product. Finally, the grafting rate is calculated according to the following formula (2).

$$GD = \frac{(V_{BLACK} - V_1) C_{HCL} \times M}{2m \times 1000} \times 100\%$$

2

Formula:

C_{HCL} —the concentration of isopropanol hydrochloride consumed (mol/L)

V_{BLACK} —volume of hydrochloric acid consumed in the blank test (mL)

V_1 —Volume of hydrochloric acid consumed in V_1 test (mL)

m— mass of the sample (g)

M—molar mass of maleic anhydride (98.6 g/mol)

2.4 The Orthogonal experiments

The impact of several factors, including DCP concentration (A), MAH concentration (B), reaction time (C), reaction temperature (D), and initiator titration time (E)[16], on the properties of grafted materials were systematically evaluated through the use of orthogonal experiments, and the optimal combination of grafting reaction process was explored by partial experiments instead of comprehensive experiments, and products with high grafting rate were obtained at the same time. The orthogonal experiments and experimental results are listed in Tables 2 ,3 and 4.

Table 2
Table of orthogonal factors

Factors	A(wt%)	B(wt%)	C(h)	D(°C)	E(min)
levels					
1	1	15	1.5	120	10
2	2	20	2	130	20
3	3	25	2.5	140	30
4	4	30	3	150	40

A: DCP concentration; B: MAH concentration; C: reaction time; D: reaction temperature; E: initiator titration time.

Table 3
Table of orthogonal experiments

Number	A	B	C	D	E	GD
	(wt)%	(wt)%	(h)	(°C)	(min)	%
1	1	15	1.5	120	10	1.47
2	1	20	2	130	20	2.57
3	1	25	2.5	140	30	3.14
4	1	30	3	150	40	2.70
5	2	15	2	140	40	1.70
6	2	20	1.5	150	30	2.56
7	2	25	3	120	20	2.33
8	2	30	2.5	130	10	2.45
9	3	15	2.5	150	20	1.94
10	3	20	3	140	10	2.70
11	3	25	1.5	130	40	2.50
12	3	30	2	120	30	2.57
13	4	15	3	130	30	3.55
14	4	20	2.5	120	40	4.11
15	4	25	2	150	10	3.19
16	4	30	1.5	140	20	4.19

Table 4
Orthogonal data processing results

Factors	A	B	C	D	E
range					
ki1	2.470	2.165	2.680	2.620	2.453
ki2	2.260	2.985	2.507	2.768	2.758
ki3	2.428	2.790	2.910	2.933	2.955
ki4	3.760	2.978	2.820	2.597	2.753
R	1.500	0.820	0.403	0.336	0.502

Upon examination of the data in Table 4, it can be seen that the concentration of the initiator has the greatest impact on the rate of grafting in the product, while the temperature of the reaction bears the least

influence on this rate.

The influence of each factor on the grafting rate of products is: A B E C D

From the k value in Table 4, the optimal grafting process conditions with DCP as the initiation system can be obtained: $A_4B_2C_3D_3E_3$

DCP concentration (4%), MAH concentration (20%), reaction time (2.5h), reaction temperature (140°C), initiator titration time (30min). The grafting rate at this time is(4.35%).

2.5 Optimization of experimental conditions

2.5.1 Effect of DCP concentration on grafting rate

From Tables 3 and 4, as the DCP concentration increases, the grafting rate also enhances and reaches its peak value of 100% at 4% DCP concentration.

The results of the experiment indicate that as the concentration of DCP rises, so does the rate of grafting. If a small quantity of DCP is present, the macromolecular radical concentration is low, leading to a reduced likelihood of colliding with the grafted monomer, leading to a low grafting rate of the resulting product. However, as the quantity of DCP increased, the grafting rate of the product rapidly rose. This was caused by a corresponding increase in macromolecular radical concentration along with the concentration of primary radicals. The outcomes in the table demonstrate this, when the concentration increases from 3–4%, the product grafting rate increases rapidly [13, 17].

2.5.2 Effect of MAH concentration on grafting rate

From the orthogonal test analysis results, it can be seen that when the dosage of MAH is 20%, the grafting rate reaches the highest, the grafting rate of the grafted product depends not only on the concentration of free radicals in the PE macromolecular chain, but also on the concentration of MAH[16]. Low MAH concentrations result in a decrease of grafting rate because there is less probability of collision between grafted monomers and macromolecular radicals. As the concentration of MAH increases, the possibility of collision between molecular radicals and MAH also increases. Consequently, the rate of grafting continues to rise [18–20]. When the MAH concentration is too high, the chance of collision with the DCP radical increases, consuming part of the MAH and DCP. This not only reduces the DCP efficiency, but also reduces the amount of grafting monomers to undergo grafting reactions, leading to a decrease in the grafting rate.

2.5.3 Effect of reaction time on grafting rate

It is evident from Table 3 that a short reaction time results in a low grafting rate, whereas an extended reaction time can lead to a substantial increase in the grafting rate [21]. In the case of too short reaction time (1-2h), the collision probability between PE macromolecular chains forming macromolecular radicals and MAH molecules is relatively small, and the grafting rate is low because the grafting reaction is not complete and the specified reaction time has not been reached. If the reaction time is in the range of 2 to 3 hours, the grafting rate exhibits a rapid increase due to the ample time that macromolecular free radicals and MAH molecules have to collide, which results in a gradual completion of the reaction. Consequently, the grafting rate exhibits a marked surge during this time period.

2.5.4 Effect of reaction temperature on grafting rate

The data presented in Table 3 unmistakably illustrates that at lower reaction temperatures, there is a corresponding lower grafting rate. As the reaction temperature falls within the range of 120°C to 130°C, the grafting rate experiences a rapid and substantial increase. Once the temperature reaches 140°C, the grafting rate reaches its peak, after which, further temperature increases result in a gradual decline in the grafting rate.

First of all, temperature has a large influence on the decomposition rate of the initiator. At excessively low reaction temperatures, the DCP's decomposition rate is sluggish, leading to diminished reactivity. In other words, the macromolecular free radicals formed are few, so that the collision probability between macromolecular radicals and grafted monomers is small. Conversely, as the temperature increases, the decomposition rate of DCP escalates, enhancing the likelihood of collisions between the reactive macromolecular radicals and MAH, so the grafting rate increases. However, when the reaction temperature is too high, the crosslinking reaction intensifies, resulting in a decrease in grafting rate [13]. Secondly, high temperatures have a negative effect on the initiation efficiency of the DCP. It can also reduce the grafting rate of the product [22].

2.5.5 Effect of initiator titration time

The DCP addition rate has a significant effect on graft polymerization as shown in Table 3. Initiator titration promotes a higher grafting rate as compared to one-time addition due to the sustained presence of initiator. The method of adding the initiator, which is done dropwise, results in a controlled distribution of macromolecular radicals. Thus, preventing an overload of free radicals in the reaction system, thereby ensuring a smooth and efficient graft polymerization. When the drop-adding time is greater than 30min, the grafting rate hardly increases.

3. Analysis of grafted products

3.1 Molecular structure characterization of grafted products

Figure 4 illustrates the additional set of absorption peaks that occur in the infrared spectrum following PE grafting. Wavelengths of (1785.73 cm^{-1} , 1771.50 cm^{-1} , 1373.31 cm^{-1} , 1860.91 cm^{-1} , 1023.93 cm^{-1} , and 917.83 cm^{-1}) are among these peaks. In addition, two distinctive absorption peaks at wavelengths of 1785.73 cm^{-1} and 1771.50 cm^{-1} , corresponding to symmetric and asymmetric ($\text{C}=\text{O}$) telescopic vibrations of pentacyclic anhydride, are present in the reaction products, namely Vas ($\text{C}=\text{O}$) and Vs ($\text{C}=\text{O}$). Grafted map has an absorption peak at 1465.26 cm^{-1} , which corresponds to the typical absorption peak of the carboxyl group ($\text{C}=\text{O}$) in MAH [23, 24]. Meanwhile, the characteristic peak at 1023.93 cm^{-1} belongs to the telescopic vibration peak of the symmetric ring VS ($=\text{C}-\text{O}-\text{C}=\text{O}$). At a wavelength of 917.83 cm^{-1} , a new absorption peak is noticeable. This peak is also the symmetrical ring telescopic vibration peak of saturated five-membered cyclic anhydride [25–27]. In summary, these new absorption peaks can indicate that maleic anhydride is grafted onto PE macromolecules.

3.2 Crystallization properties of grafted copolymers

As can be seen from Fig. 5, the melting point of PE and modified PE is between 120°C and 130°C , and the grafted PE is slightly reduced. Engrafted PE has a tendency to concave crystallization peaks when melting at 110°C to 120°C , while modified PE is relatively smooth, because after MAH grafting, nucleation sites are added on the surface of PE, which can make modified PE crystallize rapidly, thereby making the cold crystallization peak disappear. Based on the examination of the size and position of the endothermic peak depicted in Fig. 6, it can be deduced that the polymer performance improves as thermal stability decreases. Additionally, the curve tends to be flatter within a specific temperature range. Obviously, the grafted PE stability is better.

Calculate the crystallinity of the sample to be measured according to Eq. (3):

$$X_c = \frac{\Delta H_m - \Delta H_{CC}}{\Delta H_m^0 \times f} \times 100\% \quad (3)$$

In Eq. (3), X_c is the crystallinity of the sample (%); ΔH_m is the melting enthalpy value of the sample ($\text{J}\cdot\text{g}^{-1}$); ΔH_{CC} is the cold crystal enthalpy of the sample ($\text{J}\cdot\text{g}^{-1}$); ΔH_m^0 is the melting enthalpy value of PLA at 100% crystallinity ($293\text{ J}\cdot\text{g}^{-1}$); f is the mass fraction of PLA (wt %) [28].

The results were as follows: the X_c of PE was 35.33% for PE and 36.90% for PE after grafting modification, and the experimental results showed that g-MAH grafting promoted the crystallization of PE, which was due to the synergy of heterogeneous nucleation and plasticization of g-MAH. With the development of the crystallinity of PE-g-MAH, which leads to the dwindle in their melting point and a rise in their melting enthalpy.

3.3 XRD analysis of products before and after grafting

Table 5
Crystal structure parameters of PE and PE-g-MAH

Samples	Diffraction on peak	$2\theta(^{\circ})$	$d(\text{\AA})$	$\beta_0(^{\circ})$	$L_{hkl}(\text{\AA})$	$X_C(\%)$
PE	110	21.726	4.087	0.472	133	35.33
	200	24.079	3.693	0.531	133	
PE-g-MAH	110	21.813	4.082	0.527	124	36.90
	200	24.187	3.682	0.541	99	
2θ : two times the diffraction degree; d : Crystalline pitch; β_0 : corrected half-peak width; L_{hkl} : apparent thickness of crystal; X_C : the degree of crystallization						

Figure 7 and Table 5 present the diffraction angles for both PE and PE-g-MAH. The diffraction angles measured for these materials are 21.8° and 24.1° , respectively. Noticeably, there are two distinct diffraction peaks that can be observed, and they align with the crystal planes [110] and [200] of the monoclinic crystal system of PE. When we compare the crystal structure parameters listed in the table for both PE and PE-g-MAH, it becomes clear that the inclusion of maleic anhydride has a negligible effect on the spacing between these crystal planes. However, it does result in a reduction in the apparent thickness of polyethylene crystals within these planes. Analyzing the diffraction patterns displayed in Fig. 7, it is apparent that the introduction of maleic anhydride brings about significant changes. The positions of the two diffraction angles shift noticeably, and both the peak area and peak width undergo substantial increases. The changes in other places are very small, indicating that the maleic anhydride on the grafting does not change the crystal structure of polyethylene, nor does it change the crystal form. Compared to PE, the crystal plane spacing of PE-g-MAH varies only slightly. Moreover, the incorporation of MAH elevates the surface energy of PE while concurrently diminishing the structural regularity of the polyethylene matrix. Comparing the data before and after grafting in the table, it can be seen that while the grafting process improves the grafting rate, the crystallinity of PE is also improved.

4. Conclusions

To sum up, according to the guidance of solubility parameters and the simulation and verification of HSP experiments, xylene is obtained as the most economical solvent for recovery of PE. Upon conducting the orthogonal test, it was found that the optimal conditions for graft polymerization were: DCP concentration (4%), MAH concentration (20%), reaction time (2.5h), reaction temperature (140°C), initiator titration time (30min). Under optimal conditions, the MAH utilized for the grafting procedure did not alter the crystalline structure of the PE or affect its crystalline form. This is attributed by the fact that the presence of polar groups raises PE's surface energy and decreases the regularity of its structure. The temperature characteristics of bitumen can be derived from the analysis of the size and position of the endothermic peak in the DSC graph. The modified PE has better stability. The temperature characteristics

of asphalt are similar to a certain extent. This is of great significance for subsequent asphalt modification.

Declarations

CRediT authorship contribution statement

Xin Wang: Data curation, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Shenglong Yu:** Investigation, Methodology, Validation, Supervision. **Yuan Li:** Investigation, Data curation, Methodology. **Yue Fang** Writing – review & editing. **YiFan Wang:** Conceptualization, Data curation. **Nuerbiya Yalikun:** Conceptualization, Validation, Supervision, Writing – review & editing, Project administration, Funding acquisition.

Acknowledgements

Over the course of my researching and writing this paper, I would like to express my thanks to all those who have helped me. First, I would like express my gratitude to all those who helped me during the writing of this thesis. A special acknowledgement should be shown to Professor Mrs. Nuerbiya, from whose lectures I benefited greatly, I am particularly indebted to Mrs. Nuerbiya who gave me kind encouragement and useful instruction all through my writing. Thanks to the State Key laboratory of Chemistry and Utilization of Carbon Based Energy Resources, Xinjiang Key Laboratory of Coal Cleaning Conversion & Chemical Engineering.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Figures

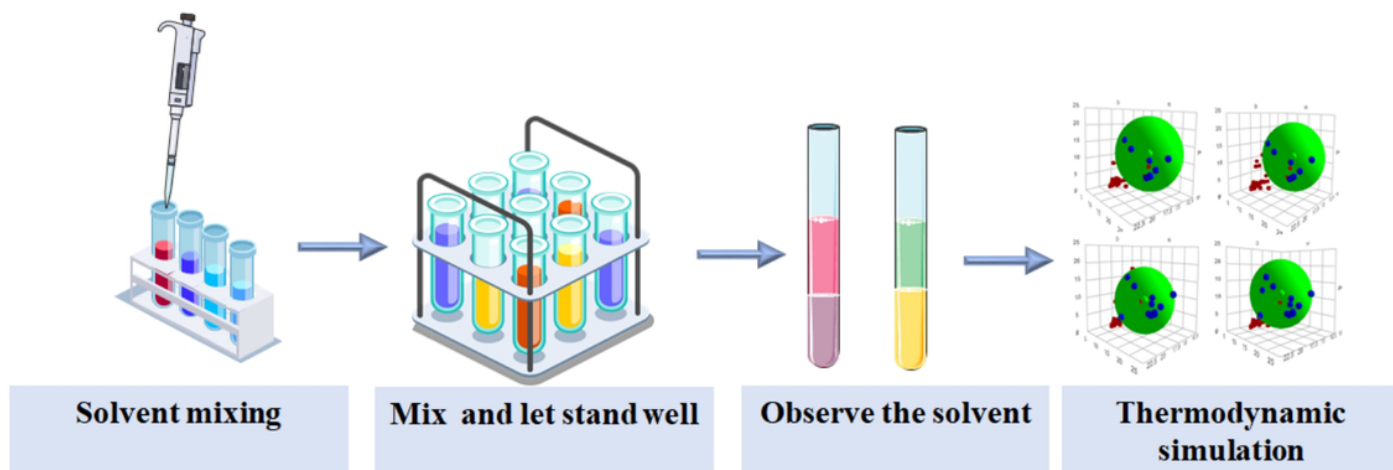


Figure 1

Schematic diagram of the HSPiP software process for determining HSPs in materials

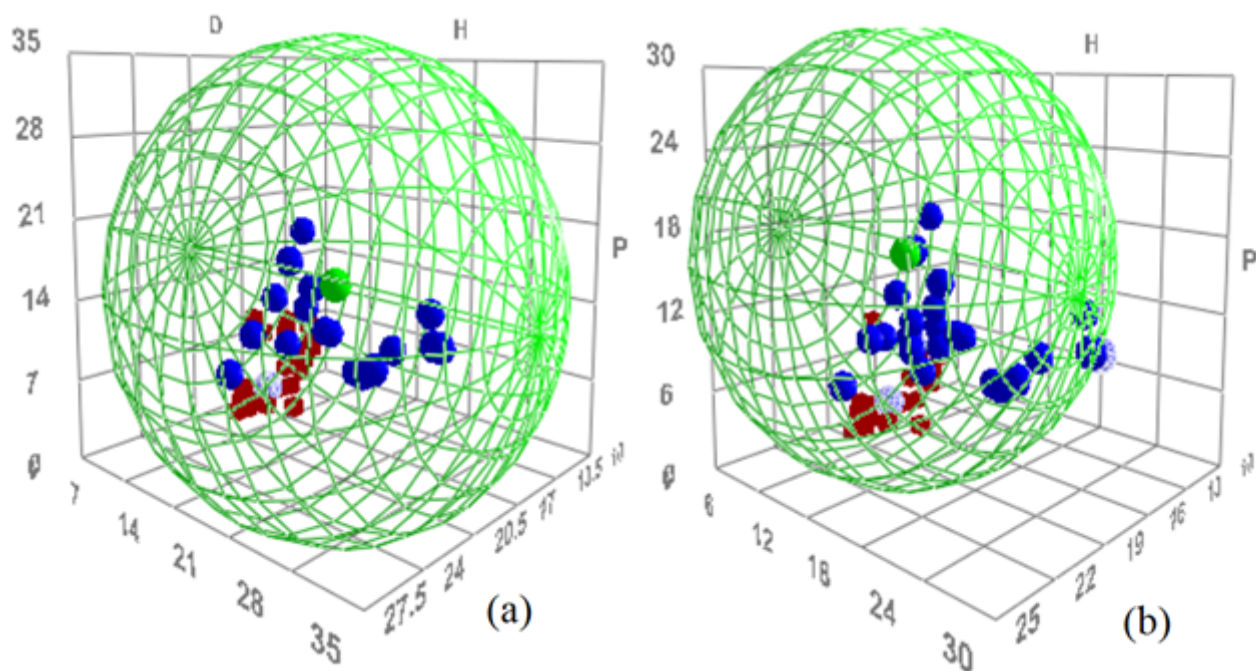


Figure 2

PE's Hansen solubility parameter ball

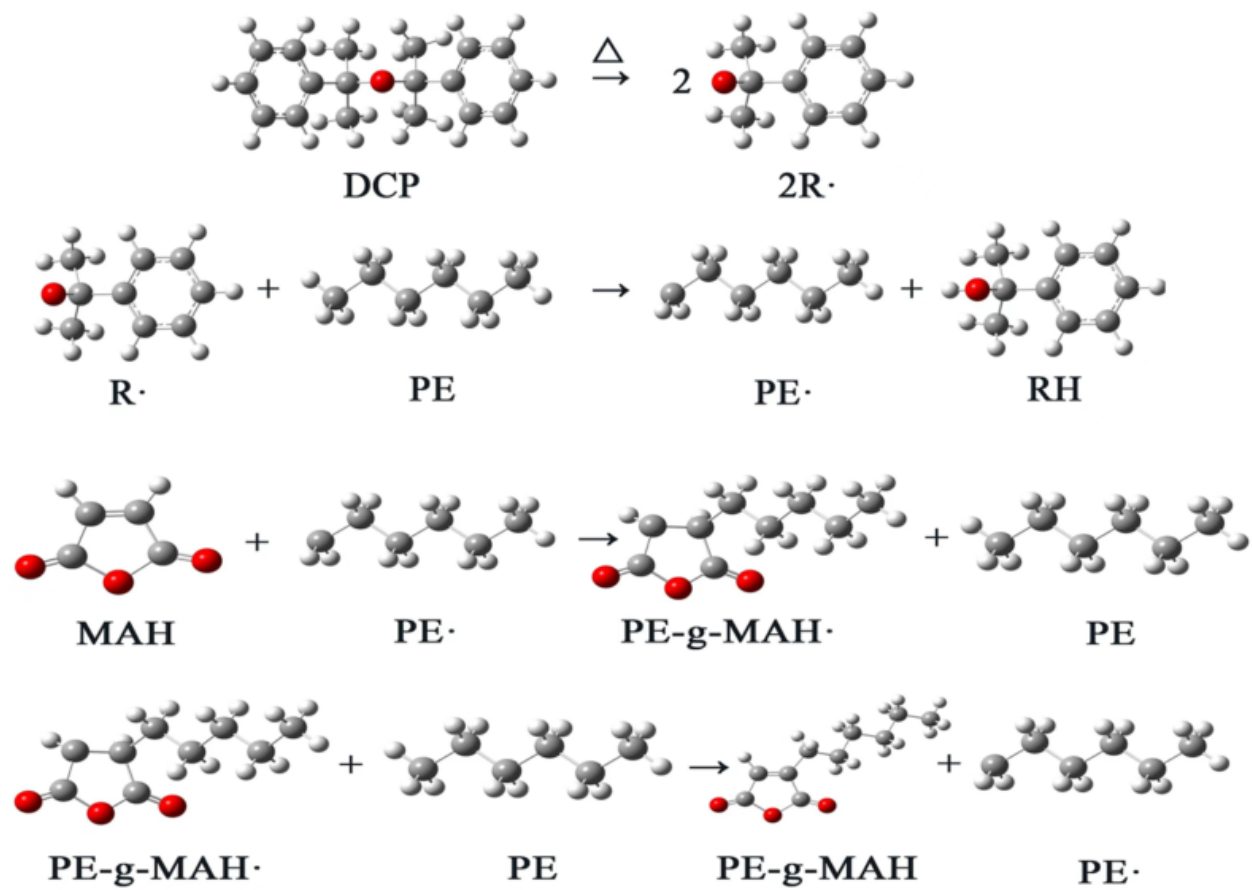


Figure 3

Grafting reaction mechanism diagram

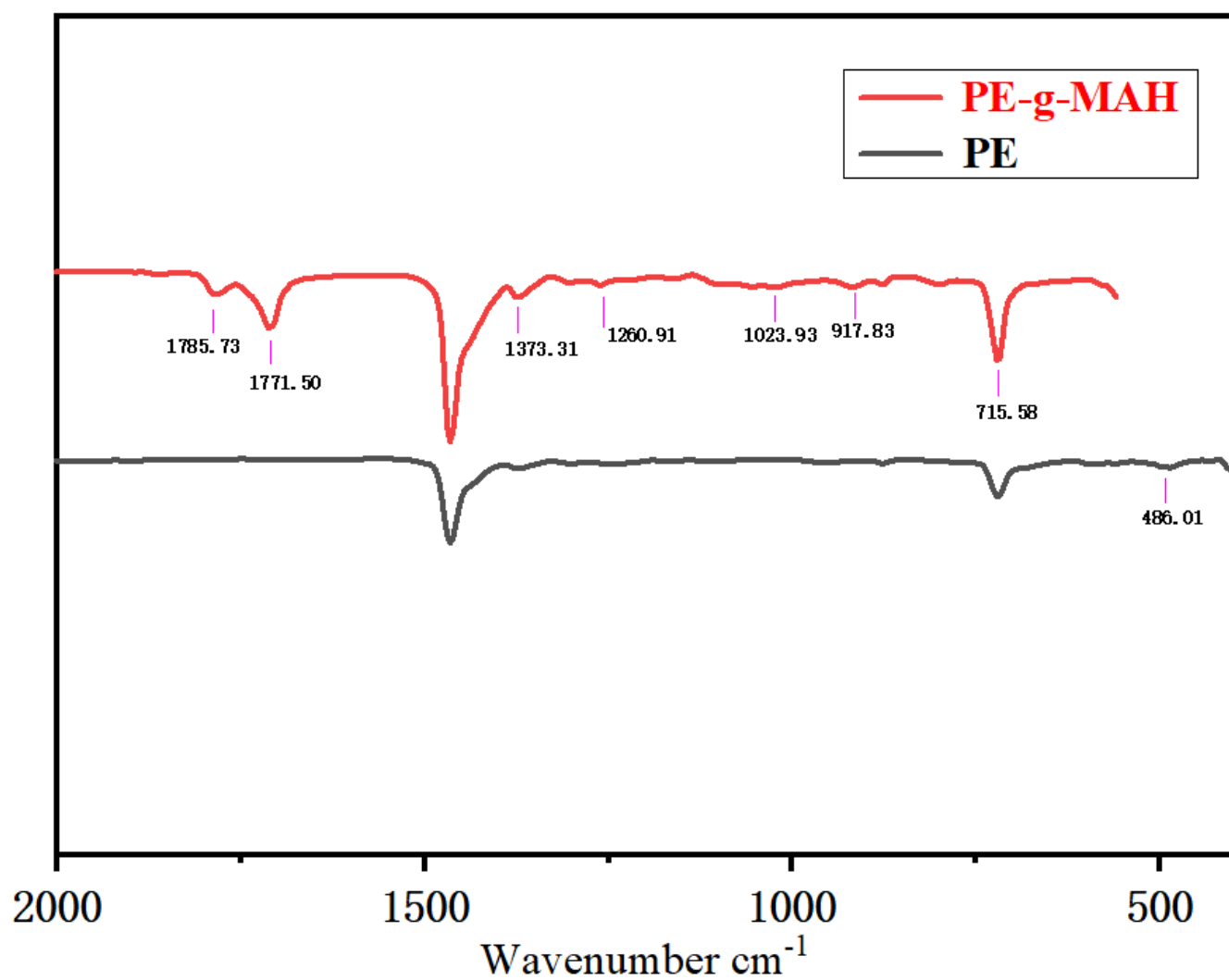


Figure 4

Infrared spectroscopy analysis of PE before and after grafting

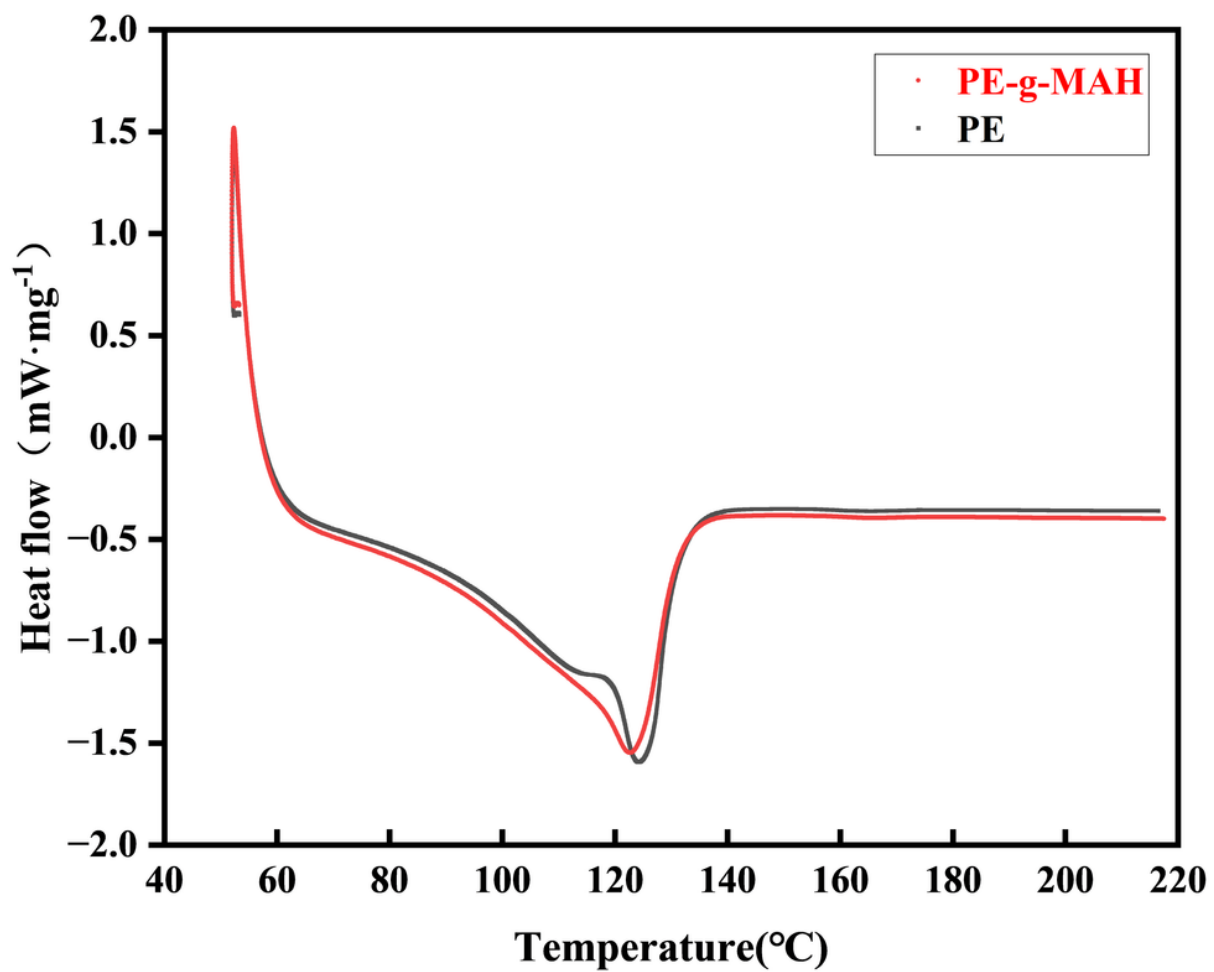


Figure 5

Melting diagram of PE and grafted PE

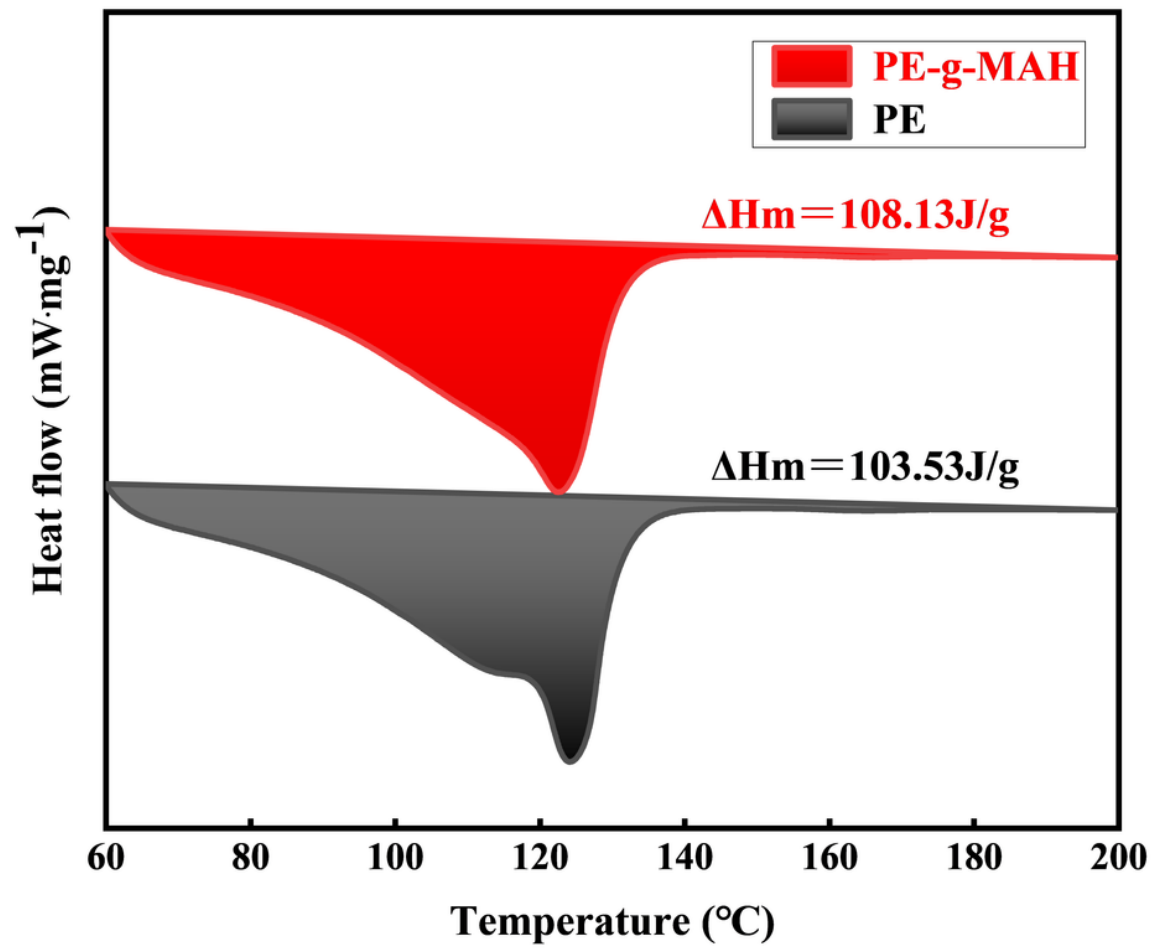


Figure 6

Diagram of the melting enthalpy of PE with grafted PE

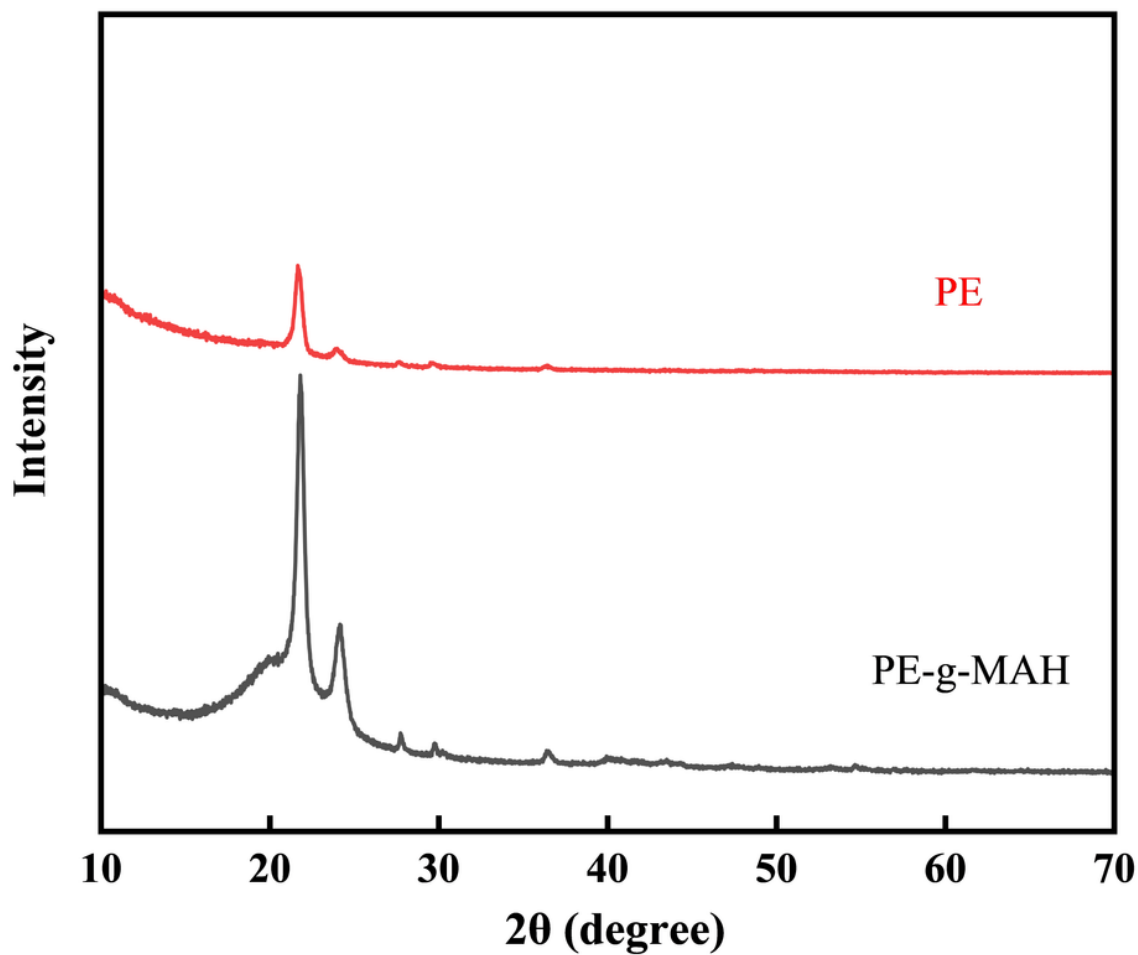


Figure 7

Wide-angle X-ray diffraction patterns of PE and PE-g-MAH

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